

Recall :

Fourier Transform of x :

$$X(\omega) = \int_{t=-\infty}^{+\infty} x(t) \cdot e^{-j\omega t} dt. \quad \leftarrow \text{Forward Fourier Transform}$$

Fourier Inversion Theorem:

If both x and X are absolutely integrable and continuous:

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(\omega) \cdot e^{j\omega t} d\omega. \quad \leftarrow \text{Inverse Fourier Transform}$$

$$x \xleftrightarrow{F} X, \quad X = F(x), \quad x = F^{-1}(X)$$

I. On the Existence of the Fourier Transform

If x is absolutely integrable, i.e., $\int_{-\infty}^{+\infty} |x(t)| dt$ is finite.

$$\begin{aligned} \left| \int_{-\infty}^{+\infty} x(t) \cdot e^{-j\omega t} dt \right| &\leq \int_{-\infty}^{+\infty} |x(t) \cdot e^{-j\omega t}| dt \\ &= \int_{-\infty}^{+\infty} |x(t)| dt \\ &< +\infty \end{aligned}$$

If x is absolutely integrable $\Rightarrow$ Fourier Transform X exists.

If x is not absolutely integrable, we usually take the approach of distributions. This will be considered later.

Facts:

If x is absolutely integrable:

1. $X(\omega)$ exists for all ω .
2. X is continuous in ω .
3. $|X(\omega)| \rightarrow 0$ as $\omega \rightarrow +\infty$ and $\omega \rightarrow -\infty$
(Riemann-Lebesgue lemma)

II. Duality:

Assume x & X are absolutely integrable and continuous.

$\Rightarrow$ Both forward and inverse Fourier transform equations are valid.

$$\Rightarrow x(\omega) = \int_{-\infty}^{+\infty} x(t) \cdot e^{-j\omega t} dt, \quad x(t) = \int_{-\infty}^{+\infty} \frac{X(\omega)}{2\pi} \cdot e^{j\omega t} d\omega.$$

Suppose $y(t) = x(t)$

What is $Y(\omega)$?

$$Y(\omega) = \int_{-\infty}^{+\infty} x(t) \cdot e^{-j\omega t} dt$$

$$Y(\omega) = 2\pi \int_{-\infty}^{+\infty} \frac{x(t)}{2\pi} e^{-j\omega t} dt.$$

$$2\pi x(t) = 2\pi \int_{-\infty}^{+\infty} \frac{X(\omega)}{2\pi} \cdot e^{j\omega t} d\omega.$$

t ω
 $\downarrow$ $\downarrow$
 ω $-t$

$$\Rightarrow \gamma(\omega) = 2\pi \gamma(-\omega)$$

In summary: If $y(t) = x(t)$

then $\gamma(\omega) = 2\pi \gamma(-\omega)$.

Duality

If $x(t) \xleftrightarrow{F} X(\omega)$, then $X(t) \xleftrightarrow{F} 2\pi \gamma(-\omega)$.
provided both γ and X are continuous and absolutely integrable.

Examples:

$$(1) \quad x(t) = e^{-a|t|}, \quad a > 0$$

$$X(\omega) = \frac{2a}{a^2 + \omega^2}$$

$$e^{-a|t|} \xleftrightarrow{F} \frac{2a}{a^2 + \omega^2}$$

We also know that both x and X are continuous and absolutely integrable

$\Rightarrow$ we can apply duality.

$$\begin{aligned} \frac{2a}{a^2 + \omega^2} &\xleftrightarrow{F} 2\pi \cdot \gamma(-\omega) = 2\pi \cdot e^{-a|-\omega|} \\ &= 2\pi \cdot e^{-a|\omega|} \\ X(t) & \end{aligned}$$

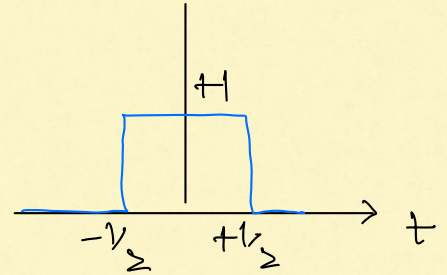
2.

Recall convolution theorem:

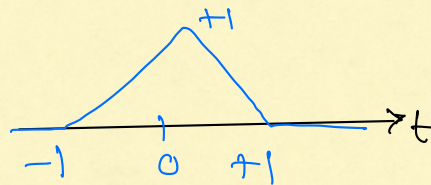
If x & h are both bounded and absolutely integrable then:

$$x * h \xleftrightarrow{F} X(\omega) \cdot H(\omega).$$

Consider $x(t) = h(t) = \Pi(t)$



$$x * h = \Delta(t)$$



$$\Pi(t) \xleftrightarrow{F} \text{sinc}\left(\frac{\omega}{2\pi}\right)$$

$$\Pi * \Pi \xleftrightarrow{F} \text{sinc}\left(\frac{\omega}{2\pi}\right) \cdot \text{sinc}\left(\frac{\omega}{2\pi}\right)$$

$$\Delta(t) \xleftrightarrow{F} \text{sinc}^2\left(\frac{\omega}{2\pi}\right).$$

Apply duality: $\text{sinc}^2\left(\frac{t}{2\pi}\right) \xleftrightarrow{F} 2\pi \cdot \Delta(-\omega) = 2\pi \Delta(\omega)$
 because Δ is even.

$$\Rightarrow \int_{-\infty}^{+\infty} \text{sinc}^2\left(\frac{t}{2\pi}\right) e^{-j\omega t} dt = 2\pi \cdot \Delta(\omega) \quad \forall \omega.$$

In particular, choose $\omega = 0$:

$$\int_{-\infty}^{+\infty} \sin^2\left(\frac{t}{2\pi}\right) dt = 2\pi.$$

$$t/2\pi = \tau \quad dt = 2\pi \cdot d\tau.$$

$$\therefore \int_{\tau=-\infty}^{+\infty} \sin^2(\tau) d\tau \cdot 2\pi = 2\pi$$

$$\Rightarrow \int_{-\infty}^{+\infty} \sin^2(\tau) d\tau = 1.$$

← important result
memorize this.

III. What if x is absolutely integrable but not continuous?

$X(\omega)$ exists $\forall \omega$ since x is absolutely integrable.

Theorem: If x satisfies the following:

1. x is absolutely integrable.

2. x has finitely many discontinuities, and each of these discontinuities is a finite gap.

$$\text{ie. } |x(t_0^-) - x(t_0^+)| < \infty$$

if x is discontinuous at t_0 .

3. $|X(\omega) \cdot \omega| \rightarrow +\infty$ as $\omega \rightarrow +\infty$, and $\omega \rightarrow -\infty$.

$|X(\omega)|$ decays like $\frac{1}{|\omega|}$ or faster as $\omega \rightarrow \pm\infty$.

Then:

$$\lim_{B \rightarrow \infty} \left[\frac{1}{2\pi} \int_{-B}^{+B} X(\omega) \cdot e^{j\omega t} d\omega \right] = \begin{cases} X(t) & \text{if } X \text{ is continuous at } t \\ \frac{X(t^-) + X(t^+)}{2} & \text{if } X \text{ is discontinuous at } t. \end{cases}$$

Detour: Principal Value (P.V.) of an Integral.

Consider an integration $\int_{-\infty}^{+\infty} f(t) dt$. The principal value

of this integration is defined to be:

$$\text{p.v.} \int_{-\infty}^{+\infty} f(t) dt = \lim_{B \rightarrow \infty} \left[\int_{-B}^{+B} f(t) dt \right]$$

←→
↓
Notation for
principal value
←→
↓
Take limit
after doing
the integration
←→
Do integration
over a finite
range -B to +B

Why consider principal value?

For some functions f $\int_{-\infty}^{+\infty} f$ might not exist, but, the principal value might still exist (examples below).

In such situations $\text{p.v.} \int_{-\infty}^{+\infty} f$ might still be useful to work with.

The previous theorem uses p.v. for such a purpose.

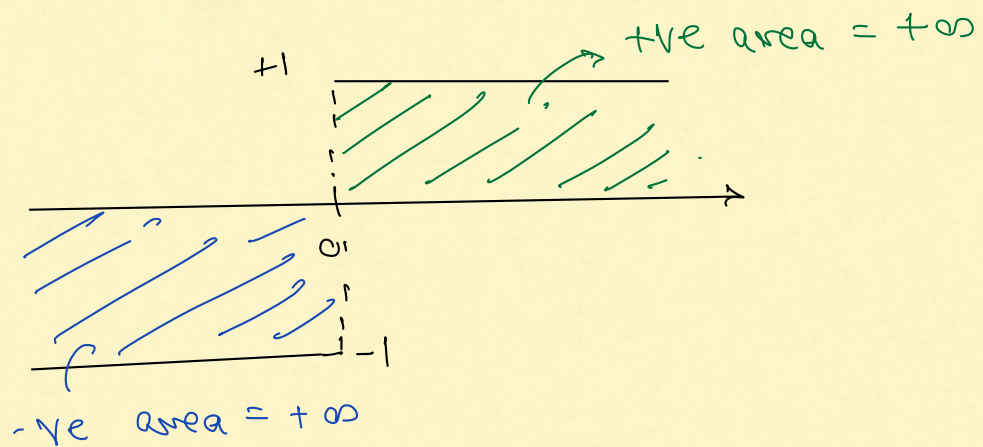
Examples for Principal Value

1. Consider $\text{sgn}(t) = \begin{cases} +1 & \text{if } t > 0 \\ -1 & \text{if } t < 0. \end{cases}$

We compare $\int_{-\infty}^{+\infty} \text{sgn}(t) dt$ and $\text{P.V.} \int_{-\infty}^{+\infty} \text{sgn}(t) dt$.

(a) $\int_{-\infty}^{+\infty} \text{sgn}(t) dt$:

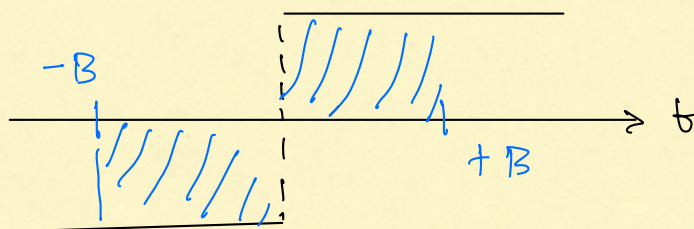
this is area under the curve. To compute this we find overall +ve area (above horizontal axis) and overall -ve area (below horizontal axis) and subtract:



Overall area = $+\infty - (+\infty)$ = undefined.

However the p.v. of this integral exists.

(b) $\text{P.V.} \int_{-\infty}^{+\infty} \text{sgn}(t) dt$:



$$\text{P.V.} \int_{-\infty}^{+\infty} \text{sgn}(t) dt = \lim_{B \rightarrow \infty} \left[\int_{-B}^{+B} \text{sgn}(t) dt \right]$$

$$\text{First find } \int_{-B}^{+B} \text{sgn}(t) dt = B + (-B) = 0$$

Then take the limit :

$$\lim_{B \rightarrow \infty} \left[\int_{-B}^{+B} \text{sgn}(t) dt \right] = \lim_{B \rightarrow \infty} 0 = 0.$$

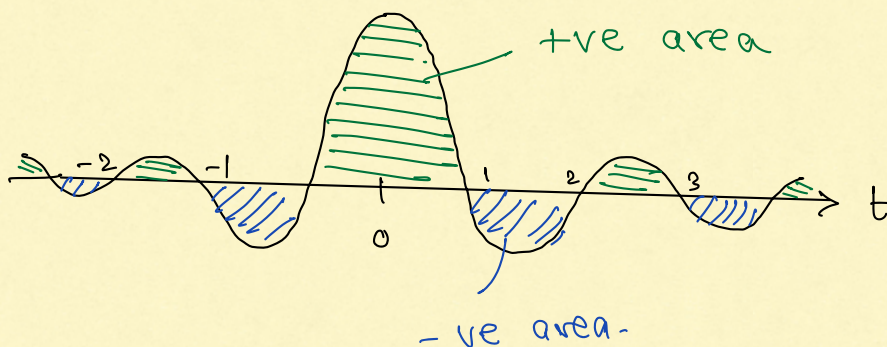
2. Second example is $\int_{-\infty}^{+\infty} \sin(t) dt$.

We will use the previous theorem to find its P.V.

$$(a) \int_{-\infty}^{+\infty} \sin(t) dt.$$

This is area under the curve

$$= \text{+ve area} - (\text{-ve area})$$



It is a known fact that for $\sin(t)$:

$$\text{+ve area} = +\infty$$

$$\text{-ve area} = +\infty$$

This is related to the fact that $\sin x$ is not absolutely integrable.

$$\begin{aligned}\text{Overall area} &= +\infty - (+\infty) \\ &= \text{undefined.}\end{aligned}$$

$$(b) \text{ p.v. } \int_{-\infty}^{+\infty} \sin(t) dt :$$

The theorem from Section III is helpful for finding this p.v.

Our starting point is $x(t) = \pi(t)$.

$$\pi(t) \xleftrightarrow{F} \text{sinc}\left(\frac{\omega}{2\pi}\right).$$

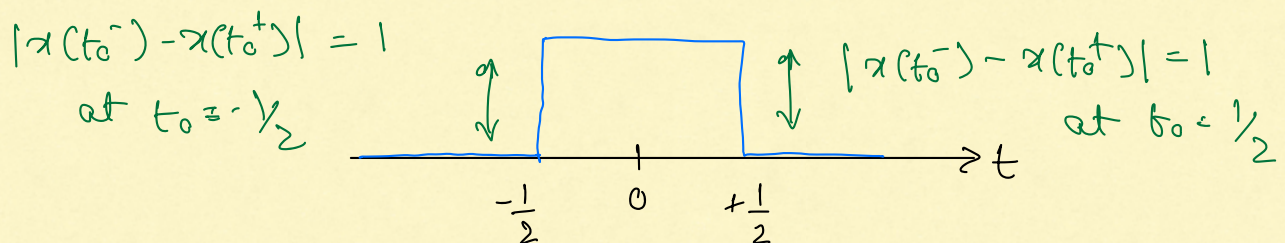
First we check if we can apply this theorem to $x(t) = \pi(t)$:

$$x(t) = \pi(t):$$

(i) $\pi(t)$ is absolutely integrable since

$$\int_{-\infty}^{+\infty} |\pi(t)| dt = 1$$

(ii) $\pi(t)$ has 2 discontinuities and these discontinuities are finite:



$$(iii) |\omega X(\omega)| = |\omega| \cdot \left| \text{sinc}\left(\frac{\omega}{2\pi}\right) \right|$$

$$|\omega X(\omega)| = |\omega| \cdot \frac{|\sin(\omega/2)|}{|\omega|/2}$$

$$= |\sin(\omega/2)| \times 2$$

$$\leq 2$$

$\therefore |\omega X(\omega)| \not\rightarrow \infty$ as $\omega \rightarrow \infty$ and $\omega \rightarrow -\infty$.

Here $|X(\omega)|$ decays as $\frac{1}{|\omega|}$.

$\Rightarrow$ We satisfy all the conditions of this theorem.

Using this theorem we have:

$$\lim_{B \rightarrow \infty} \left[\int_{-B}^{+B} \frac{1}{2\pi} \cdot \text{sinc}\left(\frac{\omega}{2\pi}\right) e^{j\omega t} d\omega \right] = \begin{cases} 1 & \text{for } -\frac{1}{2} < t < \frac{1}{2} \\ 0 & \text{for } |t| > \frac{1}{2} \\ 0.5 & \text{for } t = \pm \frac{1}{2} \end{cases}$$

$\leftarrow$ This is precisely P.V. $\int_{-\infty}^{+\infty} \frac{1}{2\pi} \text{sinc}\left(\frac{\omega}{2\pi}\right) \cdot e^{j\omega t} d\omega$.

Now consider $t=0$:

$$\text{RHS} = 1$$

$$\text{LHS} = \text{P.V.} \int_{-\infty}^{+\infty} \frac{1}{2\pi} \text{sinc}\left(\frac{\omega}{2\pi}\right) d\omega$$

Use $\tau = \omega/2\pi$, i.e., $d\tau = d\omega/2\pi$.

$$\therefore \text{LHS} = \text{P.V.} \int_{-\infty}^{+\infty} \text{sinc}(\tau) d\tau$$

Thus:

$$(a) \int_{-\infty}^{+\infty} \text{sinc}(\tau) d\tau = \text{undefined}$$

if the LHS is considered as the usual integration, i.e., area under the curve-

$$(b) \text{P.V.} \int_{-\infty}^{+\infty} \text{sinc}(\tau) d\tau = 1 \quad \leftarrow \text{important result memorize this.}$$

Important Remark

In many textbooks $\int_{-\infty}^{+\infty} \text{sinc}(\tau) d\tau$ is assumed to refer to its principal value, although they do not mention this explicitly.

Therefore it is common to find

$$\int_{-\infty}^{+\infty} \text{sinc}(t) dt = +1.$$

But remember that the LHS is actually P.V., and not the actual area under the curve.